

● RESEARCH GRADE · 99%+ PURITY · USA-BASED



RESEARCH REFERENCE & SAFETY DOCUMENTATION

BPC-157

Synthetic pentadecapeptide · Research-grade reference material

CAS

137525-51-0

FORMULA

$C_{22}H_{98}N_{16}O_{22}$

MOL. WEIGHT

1419.56 g/mol

CLASS

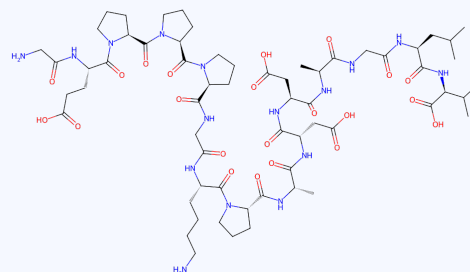
Peptide fragment

Sold for laboratory research purposes only. Not for human or veterinary use, consumption, or therapeutic application. Must be 21+ to purchase. | Independently verified · HPLC + mass spectrometry · Certificate of Analysis per lot.

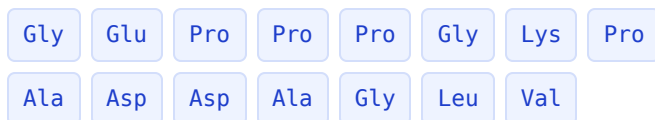
Identity & Composition

BPC-157 is a synthetic 15-amino-acid peptide corresponding to a partial sequence of a protein identified in gastric juice. Supplied as a lyophilized, research-grade reference material.

Product name	BPC-157
Synonyms	Body Protection Compound-157; PL 14736
CAS number	137525-51-0
Molecular formula	C ₆₂ H ₉₈ N ₁₆ O ₂₂
Molecular weight	1419.56 g/mol
PubChem CID	9941957
Appearance	White lyophilized powder
Solubility	Soluble in water and DMSO
Identified use	In-vitro research & development only



Molecular structure. 2D structure computed from the primary sequence; molecular formula and mass consistent with C₆₂H₉₈N₁₆O₂₂ (1419.56 g/mol).



Primary sequence (N → C): GEPPPGKPADDAGLV

Documentation note. Full hazard, handling, storage and toxicological data appear in the Safety Data Sheet section of this document. The lot-specific Certificate of Analysis (HPLC purity + mass-spectrometry identity) is provided with each order.

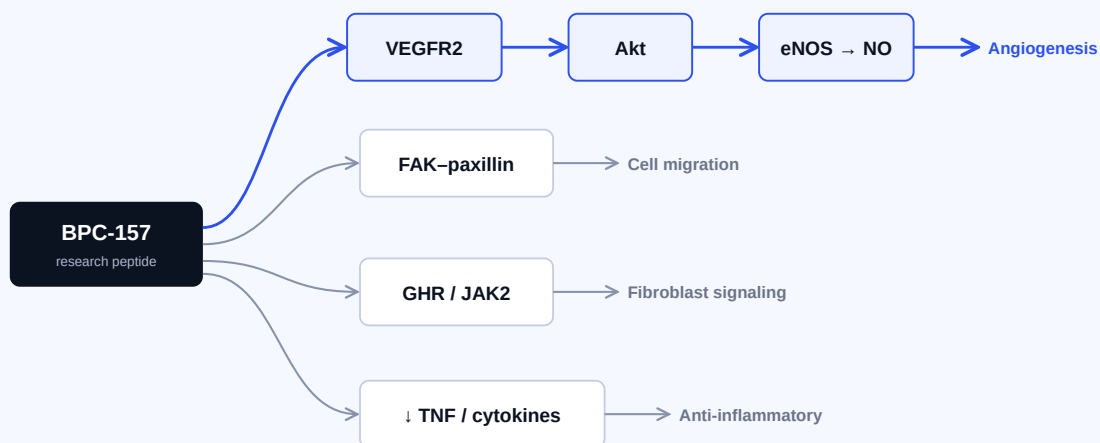
Research Background

Investigated in published preclinical research since the early 1990s across tissue-repair, gastrointestinal, vascular and neurological models. Frequently characterized as notably stable in aqueous and acidic conditions, which is one reason it has been studied across multiple routes of administration in animal models.

Mechanisms studied in preclinical models

Findings below derive from animal and cell-culture systems. They are not established effects in humans.

- **Angiogenesis via VEGFR2 signaling.** Endothelial responses reported through the VEGFR2–Akt–eNOS axis (Hsieh et al.).
- **Nitric oxide (NO) system modulation.** Interaction with the Src–caveolin-1–eNOS pathway; magnitude and direction remain under active debate in the 2025 literature.
- **Growth-hormone receptor (GHR) upregulation.** Increased GHR expression and downstream JAK2 signaling in tendon fibroblast cultures.
- **FAK–paxillin and ERK1/2 signaling.** Implicated in cell-migration and tissue-repair models.
- **Anti-inflammatory modulation.** Reduced pro-inflammatory cytokines in injury models.



Proposed pathways investigated in preclinical (animal / in-vitro) models. Schematic only; not a claim of established human effect.

What the Literature Supports — and What It Doesn't

The preclinical mechanistic literature is substantial and, in several pathways, coherent across laboratories. Human data remains very limited. Honest characterization matters for a research reference.

Composition of the reviewed literature

Vasireddi et al. (2025) systematic review, HSS Journal — 544 articles screened (1993–2024), 36 included:



■ Preclinical (animal / in-vitro): 35 ■ Clinical: 1 (a 2024 IV tolerance pilot — not an efficacy trial)

- The overwhelming majority of data is from **animal and cell-culture models**, not humans.
- Human data is limited to **small pilot studies**, primarily tolerability rather than efficacy, without independent replication.
- **No large, controlled human efficacy trials** have been published.

Appropriate characterization: a peptide with an extensive preclinical research literature and coherent proposed mechanisms, awaiting rigorous human study. Statements that BPC-157 “heals” or “treats” any condition in humans are not supported by current published evidence.

Regulatory & status notes

US FDA	Not approved for any use. Subject to FDA attention in the compounding context (identified as raising safety concerns for compounding use).
WADA	Included on the World Anti-Doping Agency Prohibited List.
Supply basis	Research reagent for in-vitro laboratory use only; not for human or veterinary administration.

Selected references

1. Vasireddi N, et al. Emerging Use of BPC-157 in Orthopaedic Sports Medicine: A Systematic Review. HSS Journal. 2025. (PMC12313605)
2. Józwiak M, et al. Multifunctionality and Possible Medical Application of the BPC 157 Peptide — Literature and Patent Review. Pharmaceuticals. 2025;18:185.
3. Hsieh MJ, et al. BPC-157 enhances angiogenesis via the VEGFR2–Akt–eNOS pathway (preclinical). PubMed 27847966.
4. Cushman CJ, et al. Local and Systemic Peptide Therapies for Soft Tissue Injury and Wound Healing. 2024. (PMC11426299)

Note. This Safety Data Sheet is compiled from public sources and weight-of-evidence assessment. Independent review by a qualified chemical-safety professional is recommended prior to use. Supplier identification and emergency-contact details must be completed by the supplier.

1 Identification

Product	BPC-157
CAS	137525-51-0
Formula / MW	C62H98N16O22 / 1419.56
Use	In-vitro research only
Restriction	Not for human/veterinary, food, drug, cosmetic, clinical, or therapeutic use
Supplier	[Supplier entity, address, phone, email]
Emergency	[24/7 monitored contact]

2 Hazard Identification

Not classified based on currently available data. Limited toxicological data; absence of classification is not a determination of absence of hazard.

Signal word: None. **Pictograms:** None required.

Precautionary: P261 avoid breathing dust/aerosols; P264 wash thoroughly after handling; P280 wear gloves/eye protection; P501 dispose per regulations.

3 Composition

Single-substance product. BPC-157 (CAS 137525-51-0), >98% research grade. Residual synthesis reagents, solvents and counter-ions may be present consistent with research-grade material. See lot COA for impurity profile.

4 First-Aid Measures

Eyes: rinse with water several minutes. **Skin:** wash with soap and water; remove contaminated clothing. **Inhalation:** move to fresh air. **Ingestion:** rinse mouth; do not induce vomiting unless directed. **Physician:** treat symptomatically; no specific antidote known.

5 Fire-Fighting

Media: CO₂, dry chemical, foam, or water spray. Combustion may produce CO, CO₂ and nitrogen oxides. Use SCBA and full protective gear.

6 Accidental Release

Avoid dust formation; ensure ventilation; use PPE per §8. Do not allow entry into drains or waterways. Collect avoiding dust; place in closed labeled container for disposal per §13.

7 Handling & Storage

Handle as bioactive peptide of unknown toxicity; minimize exposure. Weigh/reconstitute in fume hood. Store lyophilized solid tightly closed, protected from light/moisture/heat; -20 °C long-term, 2–8 °C short-term working stock; keep desiccated. Aliquot reconstituted solutions; store frozen, protected from light. Observe COA retest date.

8 Exposure Controls / PPE

No established OEL; control to ALARA. Local exhaust or fume hood for dry powder. Nitrile gloves (double-glove for neat powder); ANSI Z87.1 eye protection; lab coat; N100/P100 respirator per 29 CFR 1910.134 where controls insufficient. Eyewash/safety shower per ANSI Z358.1.

9 Physical & Chemical Properties

State	Lyophilized powder
Appearance	White powder
Odor	Odorless
Solubility	Water; DMSO
MW	1419.56 g/mol
MP/BP/Flash	Not determined

10 Stability & Reactivity

Stable under normal storage. Avoid heat, moisture, sunlight, UV, open flame. Incompatible with strong oxidizers, strong acids/bases, strong reducing agents, trace transition-metal ions. Decomposition: CO, CO₂, NO_x. No hazardous polymerization.

11 Toxicological Information

Not fully characterized. No LD₅₀/LC₅₀ available. Skin/eye: treat as potentially irritating. Sensitization: cannot be ruled out. Mutagenicity, carcinogenicity, reproductive toxicity, STOT, aspiration: no authoritative substance-specific data; not classified based on available data; hazards cannot be excluded. Avoid exposure if pregnant, nursing, or planning pregnancy.

12 Ecological Information

No substance-specific ecotoxicity, persistence or bioaccumulation data identified. Avoid release to surface water, soil, drains or sewers.

13 Disposal

Dispose per applicable local/state/federal regulations; US per 40 CFR 261–270 (RCRA). Do not dispose into sewers/waterways. Use a licensed waste disposal company.

14 Transport Information

Not regulated as dangerous goods under DOT (49 CFR), IATA or IMDG based on current classification. UN number: not applicable. Shipper must independently verify requirements.

15 Regulatory Information

US TSCA: may qualify for R&D exemption (40 CFR 720.36) per conditions of use. OSHA HazCom 2012 aligned with GHS Rev. 8. EU REACH: supplied for SR&D use; verify exemption. Not a legal determination of regulatory status.

16 Other Information

Revision [DATE] · Version 1.0. Independent review by a qualified chemical-safety professional is recommended prior to use. Sources: PubChem; peer-reviewed literature; OSHA HazCom 2012; GHS Rev. 8; OECD Test Guidelines. Provided without warranty; users responsible for verifying all information and compliance. For scientific R&D use only.

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